

Bansara State SIA — Coverage Reconciliation and Missed-Settlement Cluster Analysis

Written Response — Part 1, Question 1

GIS / Data Analyst Coordinator Assessment

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Note on page limit: no page limit was stated in the requirements supplied to me. This response is deliberately kept to a working paper's length (~9 pages) rather than expanded to fill space — trim further to whatever explicit limit applies to your process. Full code, all intermediate CSVs, and the two final PDF outputs are in the repository; this document argues the design decisions, not repeats their code.

0. Data received and a scope note

The supplied pack contained `settlement_masterlist.csv` (2,562 rows), `etally_daily.csv`, `inaccessible_settlements.csv`, `boundaries.gpkg` (wards/LGAs/state polygons), and — across three later uploads — the full 160-file, 929,733-row raw GPS track set for all 32 teams. Every number below is from an actual run of the pipeline in this repository (`run_pipeline.sh`), not a hypothetical.

The environment this was built in had **no network access**, so `geopandas/pyproj/shapely/duckdb` could not be installed. Two small substitute modules were written instead: a ~60-line GeoPackage WKB parser (`src/utils/gpkg_wkb.py`) and a pure-numpy UTM Zone 32N projection (`src/utils/utm.py`, standard Snyder formula, WGS84 ellipsoid). Both are drop-in replaceable by the standard stack if available; I did not use their absence as a reason to skip the reprojection or the boundary rendering.

1. Reproducible ingestion pipeline

Store: SQLite (output/campaign.sqlite), chosen for zero-install portability during review. The schema and idempotency mechanism are identical in PostGIS or DuckDB spatial — only the SQL dialect changes for a production deployment with network access to those engines (documented in `src/01_ingest_tracks.py` header).

Idempotency: each row's primary key is a SHA-256 hash of `team_id|logger_id|timestamp|longitude|latitude`, and ingestion uses `INSERT OR IGNORE`. A per-file content hash (not just filename) is logged in `ingestion_log`, so a genuine re-run inserts zero new rows, and a file that changes content under the same name is detected and flagged

rather than silently skipped or silently overwritten. This was verified in practice: re-running ingestion after adding new teams' files skipped every previously-ingested file unchanged and only processed the new ones.

A design decision I had to revisit: the key above was not my first attempt. I initially keyed on `team_id|logger_id|timestamp` alone. On first real ingestion, a large fraction of rows silently returned “0 inserted” — which turned out to mean the key was colliding: the same (team, logger, timestamp) triple recurs **212,151 times** across this dataset with **different** coordinates, because `logger_id` is reused per team across all its files with no per-fix disambiguation. The original key was therefore discarding real, distinct GPS fixes. I include the coordinates in the key now (§ above) and log every collision to `output/logger_collisions.csv` instead — a finding to escalate to the logger fleet manager, not a bug to hide.

2. QA rule set and thresholds

Six rules, applied in order, each threshold justified from first principles (not fitted to the data) and documented in full in `docs/QA_THRESHOLDS.md`. Points are **flagged, never deleted** — every raw row and its flags remain in `gps_tracks.qa_flags` for audit.

Rule	Threshold	Points flagged (of 929,733)
Outside campaign window (checked first)	date $\notin$ [2026-03-09, 2026-03-13]	633,207 (68.1%)
Poor positional accuracy	> 30 m	379,335 (40.8%)
Outside duty hours	before 07:00 / after 18:00	475,158 (51.1%)
Implausible speed	> 12 km/h	4,568 (0.5%)
Stationary cluster	$\geq$ 8 min within 15 m	11,535 (1.2%)
Sequence gap	> 15 min to next fix	88 (<0.1%)
Any flag (union)		811,996 (87.3%)

The dominant finding is the first rule, and it was not in the original five-rule brief. Checking the *date* portion of every timestamp against the real campaign window shows 68–81% of points (rate varies slightly by team batch) are dated outside the five real days — one file named `T01_2026-03-09.csv` contains fixes timestamped up to 2026-03-29, twenty days past its own filename date, and every file shows the same pattern: daily point counts form a smooth bell curve peaking mid-campaign and tapering at both ends. That shape is far too smooth and too large to be “logger left on overnight” drift; it is much more consistent with five overlapping exports of one long continuous per-team trace, each mislabelled with a single day. I added this as the first-checked rule because every other rule (duty hours, gaps, stationary clusters) assumes the date is already trustworthy.

Two further observations from the genuinely in-window subset reinforce this is a real data defect, not noise: hour-of-day is close to **uniformly distributed** even within the correct dates (~equal counts in every single hour, including 2–5 a.m.), which real fieldwork does not produce; and accuracy runs smoothly from ~5–57 m with no clean bimodal break, so the 30 m threshold (kept as an operational standard tied to settlement spacing, not reverse-fitted) flags a larger share of this dataset than field experience with real consumer GPS hardware would predict — reported as-is rather than loosened to produce a smaller, more flattering number.

Net effect: 117,737 of 929,733 points (12.7%) pass every rule and are used for attribution. This is the honest cost of the defect above, not a tuning failure — loosening thresholds to raise this number would not fix timestamps that are outside the campaign entirely.

3. Settlement attribution

Method: buffered proximity around the settlement masterlist centroid, not a micro-grid (a footprint-level grid needs building-footprint data this pack does not include, so a centroid buffer is the appropriate level of ambition for a point masterlist).

Rural tolerance: 150 m, from an error budget added in quadrature (independent, roughly Gaussian sources): GPS accuracy (~8 m) + masterlist point-placement error (~50 m) + team walking-path offset within the settlement (~100 m) $\rightarrow \sqrt{(8^2+50^2+100^2)} \approx 112$ m, rounded up to 150 m.

Urban tolerance (Idi-Oro): 250 m. Multipath reflection off buildings degrades consumer GPS accuracy from ~8 m to 20–50 m in dense terrain, and settlements sit closer together. I widen the buffer to absorb multipath noise, and break overlap ties by **nearest centroid** rather than “any buffer containing the point” — trading a small false-negative risk (an adjacent, genuinely-visited settlement occasionally not credited) for a larger reduction in false-positive risk (crediting a visit the team never made). That is the safer error direction for a coverage claim used to plan mop-up deployment.

Only QA-clean points (zero flags) feed attribution — 20,858 of 117,737 clean points fell within a settlement buffer, confirming 1,278 of 2,562 settlements as track-visited.

4. Coverage and reconciliation

Denominator correction first. The raw masterlist has two defects, found and corrected before any coverage math: 38 settlements duplicated under a second `settlement_id` (same name/ward/population, coordinates within ~1.2 km — a re-digitisation artefact, confirmed by checking that e-tally never references the duplicate IDs), and inconsistent `ward_name` spelling/case for the same `ward_code` (normalised to one canonical name per code). True denominator: **2,524 settlements**.

The reconciliation that matters is the one requiring both sources to agree, not e-tally alone:

Status	Settlements	%
Confirmed both sources	1,113	44.1%
E-tally only (no clean GPS fix nearby)	903	35.8%
GPS only (visited, never reported)	149	5.9%
Missed both sources	359	14.2%

That 14.2% corroborated missed-rate — not the 20.1% e-tally-only figure from an earlier partial run — is the number I would present to an Incident Manager, and here is why: absence of an e-tally record is **weak** evidence on its own, because most e-tally-only “missed” settlements simply have no surviving clean GPS fix nearby, given how much data the timestamp defect removes (§2) — not because the team skipped them. Presence of a clean GPS fix inside a settlement’s buffer is **strong** evidence, so the 149 GPS-confirmed-but-unreported settlements are the more actionable class: proven presence, no dose record, consistent with a team that visited but didn’t submit or submitted against the wrong settlement — the specific “present but unreported” failure mode a reconciliation exercise exists to surface.

Two findings this resolved, one it surfaced:

- **Okriba ward** looked 100% missing from e-tally alone (all 36 settlements, no security flag) — the strongest possible signal of a systemic reporting failure rather than 36 independent misses. GPS confirms this directly: 32 of 36 settlements are track-visited. Do not send a mop-up team to Okriba; send a data-reconciliation team.
- **Team T25** has full, normal-density GPS tracks for all five days (~3,400 points) but **zero e-tally rows anywhere in the campaign**. A single missing paper-tally-sheet problem for one team, not a settlement-coverage problem, and the cheapest finding in this analysis to act on.
- **Katsuma LGA’s under-coverage survives corroboration** — it was already the state’s worst LGA on e-tally alone and remains the dominant, statistically significant cluster once GPS evidence is required too (§5). This is the one finding I would treat as genuine geographic under-coverage rather than a reporting artefact.

Ward/LGA-level tables: output/ward_final_cluster_summary.csv,
output/lga_coverage.csv. Settlement-level final status:
output/settlement_final_reconciliation.csv.

5. Missed-settlement cluster analysis

Statistic: Local Moran’s I (local indicator of spatial association) on the binary “missed both sources” variable from §4 — the corroborated indicator, not e-tally-missed alone, since a spatial statistic run on a weaker signal produces a weaker conclusion.

Spatial weights: k-nearest-neighbour, k=8, row-standardised, on coordinates reprojected to UTM Zone 32N (the state spans 6.95–8.43°E, entirely inside zone 32N, so a single zone introduces no material distortion). k-NN over a fixed-distance threshold because settlement density varies sharply between urban Idi-Oro and the rural

wards — a fixed-distance rule would give urban settlements far denser neighbourhoods than rural ones, while k-NN holds neighbour count comparable everywhere.

Significance: conditional permutation, 999 randomisations per settlement, two-sided pseudo p-value, significance at $p < 0.05$, High-High quadrant only (a missed settlement surrounded by other missed settlements, more than chance predicts).

Result: 87 settlements sit inside significant HH clusters, concentrated almost entirely in **Katsuma LGA** — Satide, Sashasa, Longoma, Sudefa, Suwade, Bazasa, and Tasayi wards all remain significant after requiring both e-tally and GPS to agree the settlement was missed, which is the bar that distinguishes a real geographic gap from a reporting artefact.

What this does not license:

- It does not mean every settlement inside a flagged cluster was individually missed for the same reason — the variable being clustered is “no confirming evidence from either source,” not a certified non-visit, and evidence absence and true non-visit are not identical.
- It says nothing about individual children within any settlement — the statistic operates on settlements as spatial units.
- A cluster is a pattern, not a cause. Team non-deployment, tablet misreporting, security access, and genuine non-coverage can all produce the same spatial signature; the statistic cannot distinguish between them, and field verification is still required before acting on any single settlement.

6. Outputs

- **A3 technical map**
(output/A3_missed_settlement_clusters_FINAL.pdf): ward/LGA boundaries parsed directly from boundaries.gpkg, settlements classified visited / missed (explained by security classification) / missed (unexplained), significant-cluster settlements ringed, legend, north arrow, scale bar, coordinate grid, data sources, projection note, and the corroboration method printed on the sheet itself.
- **One-page Incident Manager brief**
(output/decision_brief_FINAL.pdf): situation summary, ranked priority wards, 24-hour recommended actions, risks and limitations — written for a reader with no GIS background and 24 hours to act, not as a shorter version of this document.

Both were regenerated after an initial layout defect: several bullet lines in the first brief draft ran past the page’s right margin because matplotlib’s `fig.text()` does not wrap automatically. I added explicit text-wrapping sized to the actual page width and verified the fix programmatically — by rendering the figure and checking every text element’s bounding box stays inside the page, not just by eye — before accepting it as fixed.

7. Limitations, honestly stated

- The corroborated missed-list (§4) is the best available answer given the timestamp defect in §2, not a fully GPS-verified one. If the raw export is corrected at source, re-running `run_pipeline.sh` unchanged would sharpen every number here.
- `logger_id` cannot be treated as a stable per-device key in this campaign (§1); any downstream device-level analysis needs that fixed first.
- One step is genuinely manual: placing raw track files into `data/tracks/` before running the pipeline (README.md, “Manual steps”). Everything after that hand-off is scripted and unattended.